



Cancer Epigenetics

Oral Midterm Exam I Part 2

Module I: Fundamentals of Epigenetics

JAN 11 2026

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Content

1. Background Concepts
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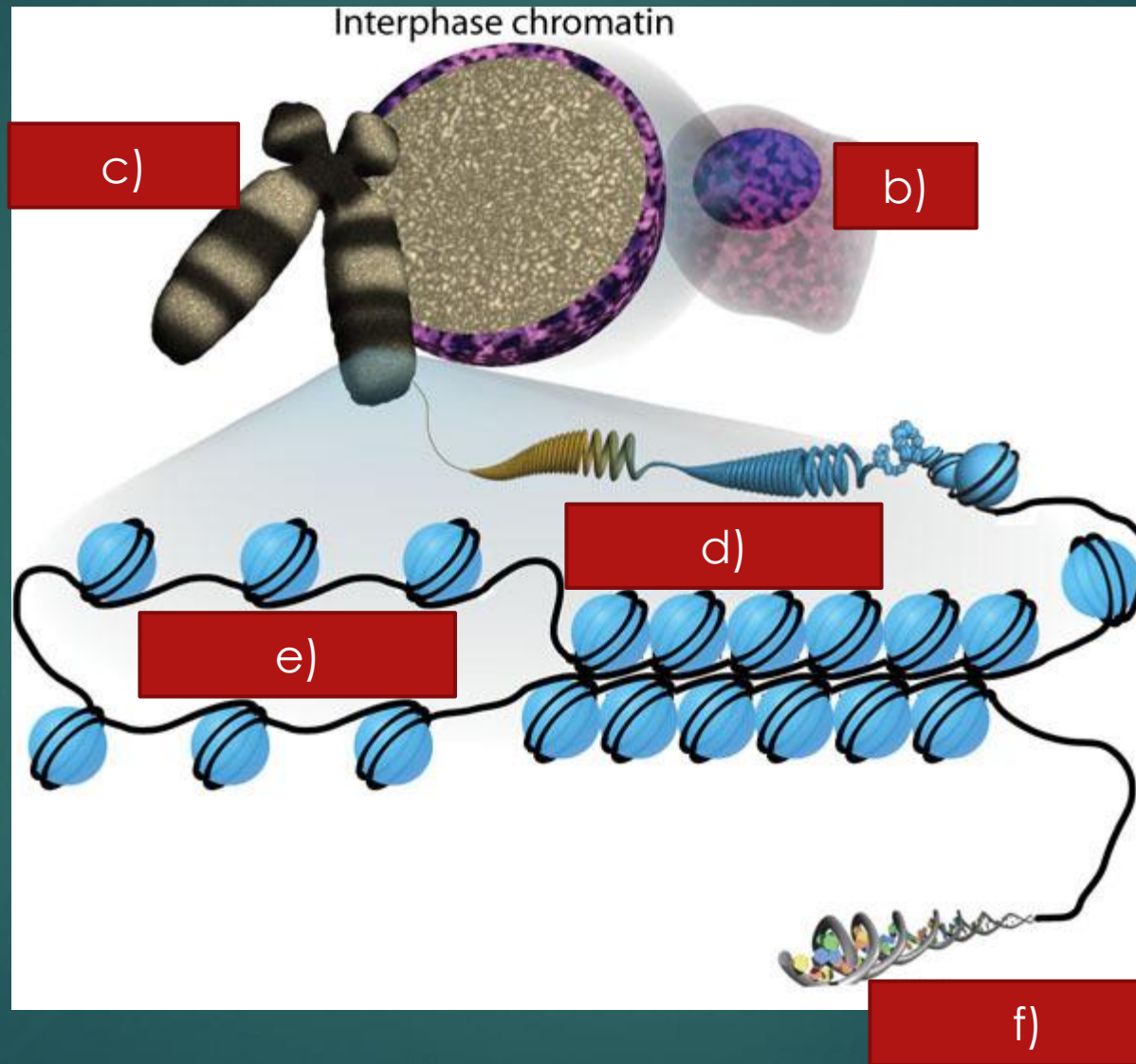
Background Concepts

?	A	A statement	B statement	B
	a	Prokaryotic transcription	The minimum level of transcription that occurs in the absence of specific activators or repressors.	1
	b	Eukaryotic transcription	It requires a specific set of general proteins that assemble at a gene promoter to recruit RNA Polymerase.	2
	c	RNAPI, RNAPII and RNAPIII	Collective name for GTFs	3
	d	The Pre-Initiation Complex (PIC)	RNAPII	4
	e	They form a platform on the DNA promoter that recruits and positions RNAPII correctly at the Transcription Start Site (TSS).	TFIIH is critical	5
	f	The crucial link between specific regulatory signals and the basal machinery.	The DNA sequence immediately surrounding the transcription start site (TSS, usually designated +1).	6
	g	The "General" Transcription Factors (GTFs)	The RNA Polymerase holoenzyme can directly recognize and bind to promoter sequences.	7
	h	TFII proteins (Transcription Factor for RNAP II)	A technique for genome-wide profiling of DNA-binding proteins, histone modifications or nucleosomes.	8
	k	Basal transcription	Require auxiliary proteins called General Transcription Factors (GTFs) to locate promoters and load the polymerase onto the DNA.	9
	l	Its activity is the most highly regulated and cell-specific of the three polymerases.	Transcribes large rRNAs, all protein-coding genes (mRNA) and many non-coding RNAs, tRNAs and small rRNAs.	10
	n	Transcription Factors (TFs)	Estrogen, Testosterone, Cortisol and Vitamin D	11
	m	ChIP-seq	Function of GTFs	12
	p	Machinery	Mediator	13
	q	It opens the DNA (helicase) and kickstarts the polymerase via phosphorylation (kinase).	The complete assembly of GTFs and RNAPII at the promoter, ready to start transcription.	14
	s	Ligands	A highly conserved set of proteins absolutely required for RNAPII transcription initiation at all core promoters.	15
	x	Core Promoter	Proteins that bind sequence-specifically to genomic DNA to control gene expression.	16
	y	Nuclear receptors	ER antagonist used in breast cancer therapy.	17
	z	Signal-Dependent Factors (Require activation)	A superfamily of ligand-inducible transcription factors.	18
	v	Tamoxifen	Nuclear Receptors, Internal Signals, Cell Surface Receptor Dependent	19

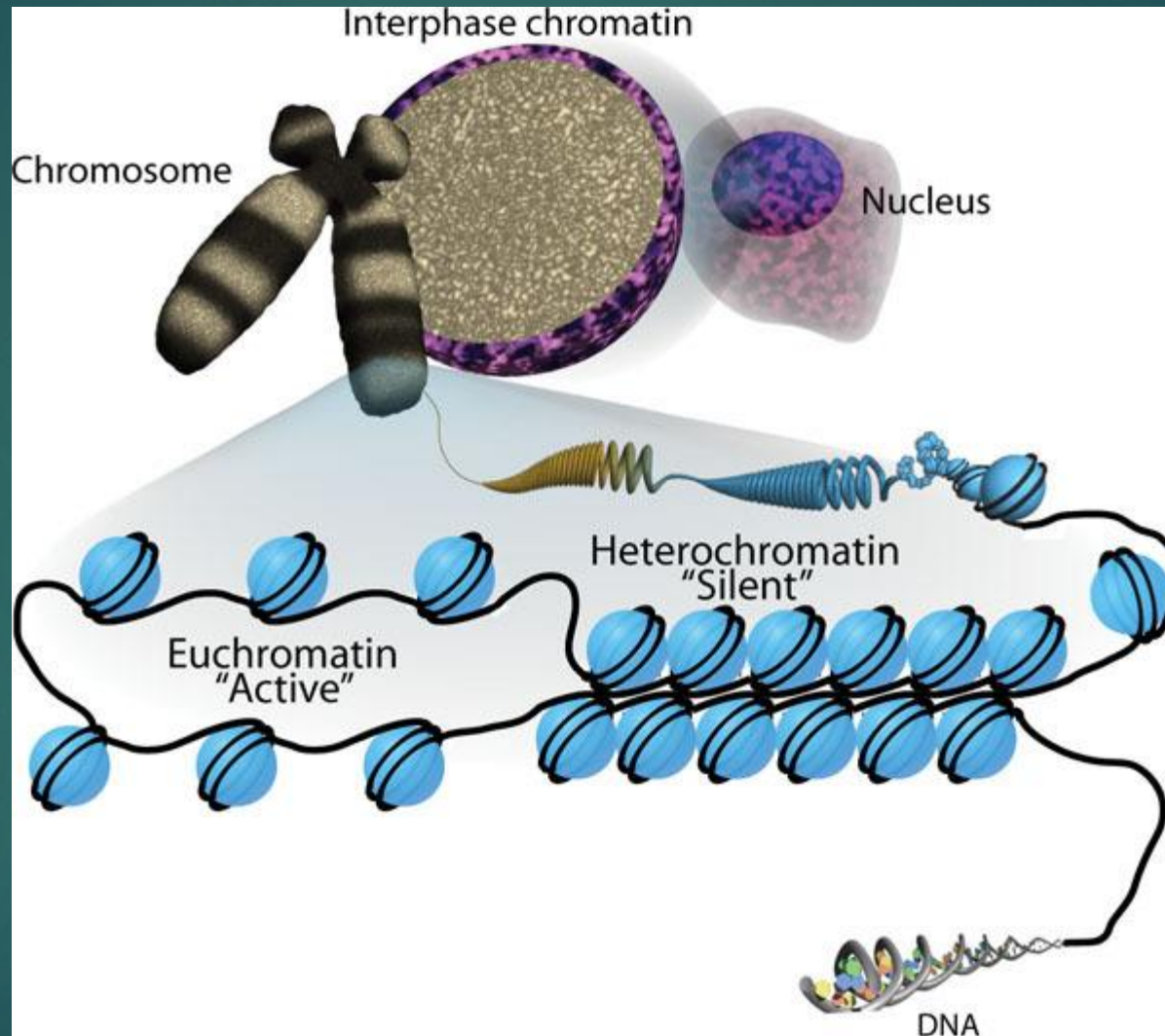
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a)

of Eukaryotic Transcription



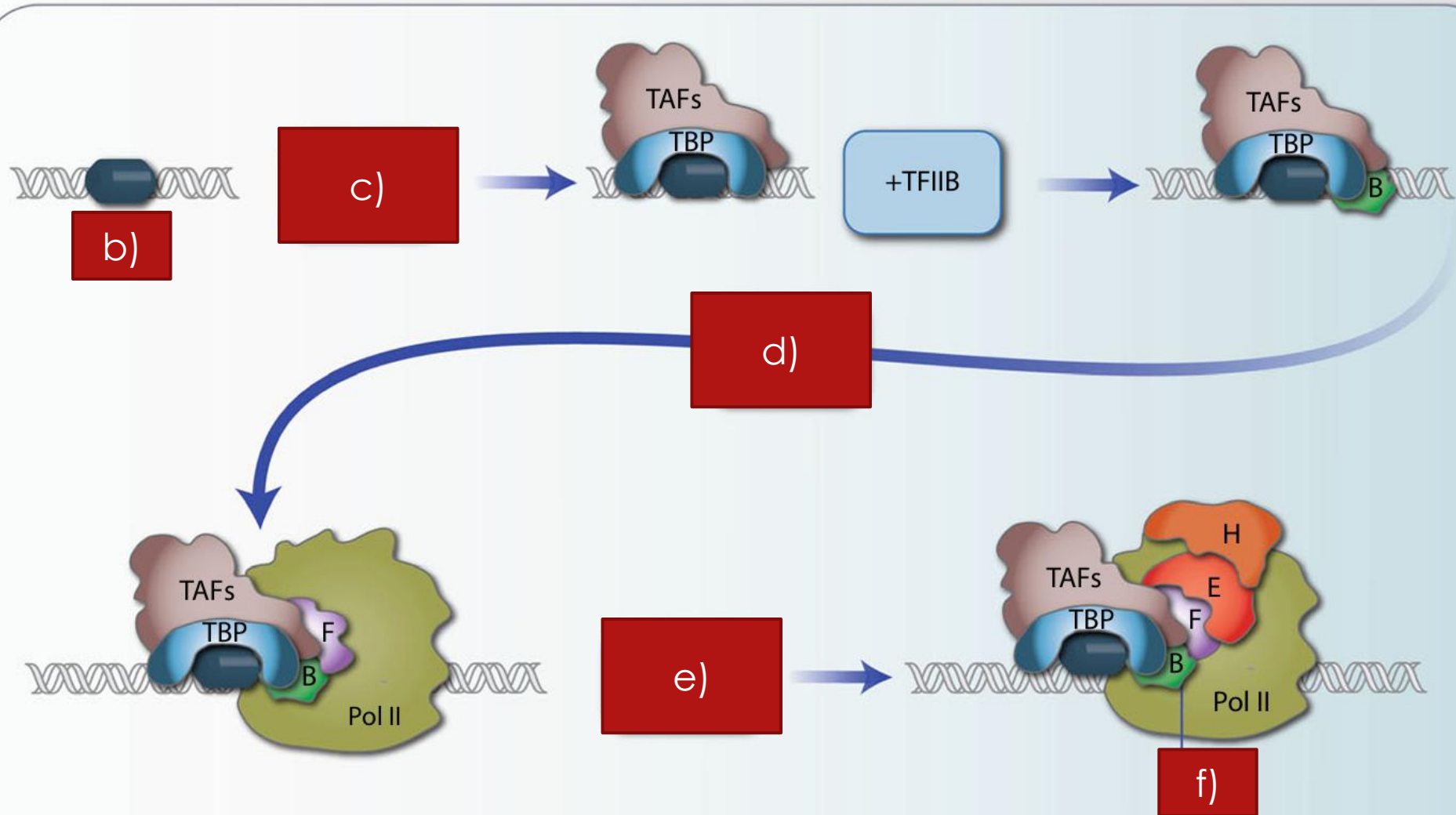
The Challenge of Eukaryotic Transcription



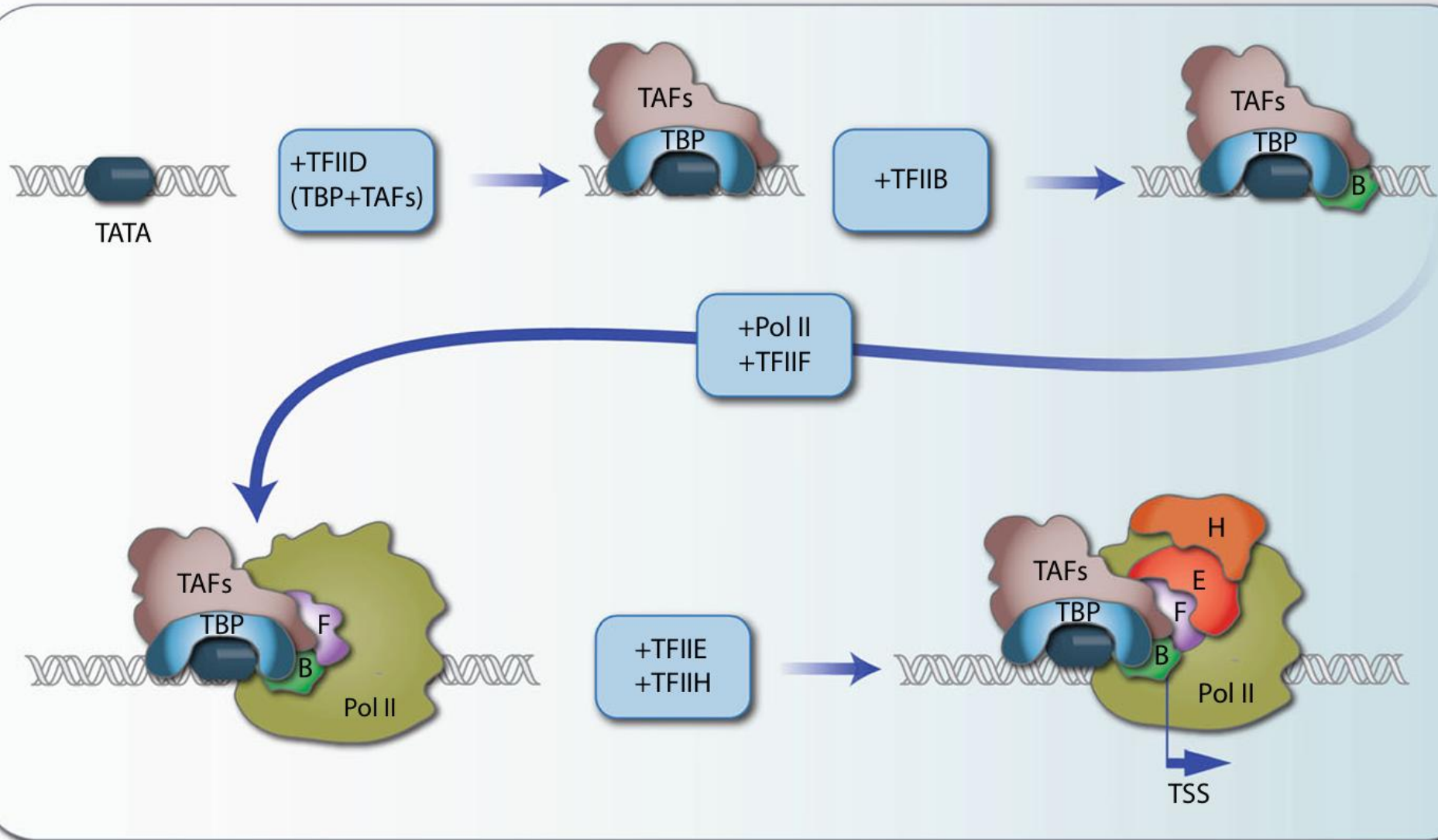
<https://www.ncbi.nlm.nih.gov/books/NBK27041/figure/thechromatinsignature.F1/>

Assembly of the

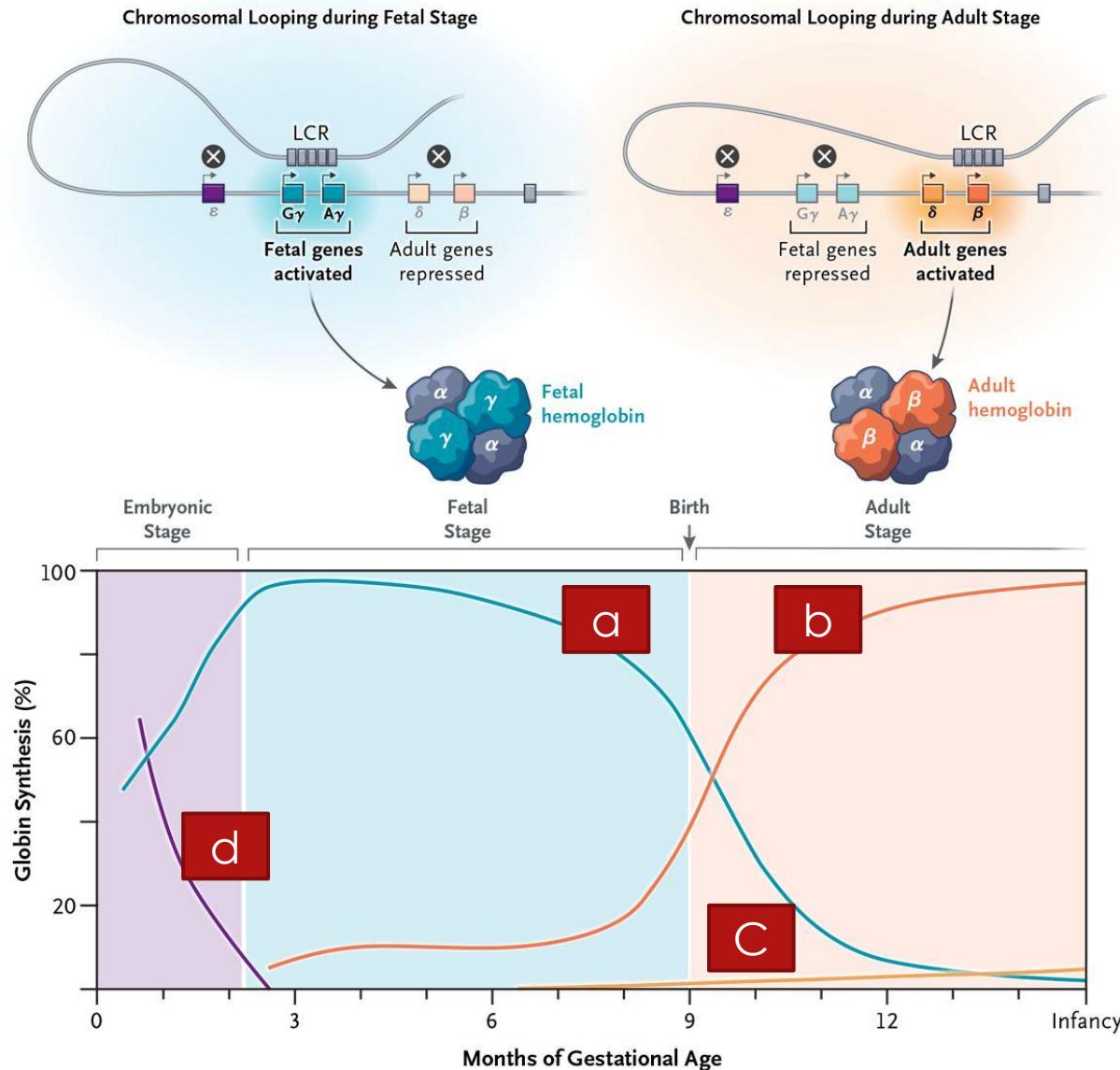
a)



Assembly of the basal transcriptional machinery

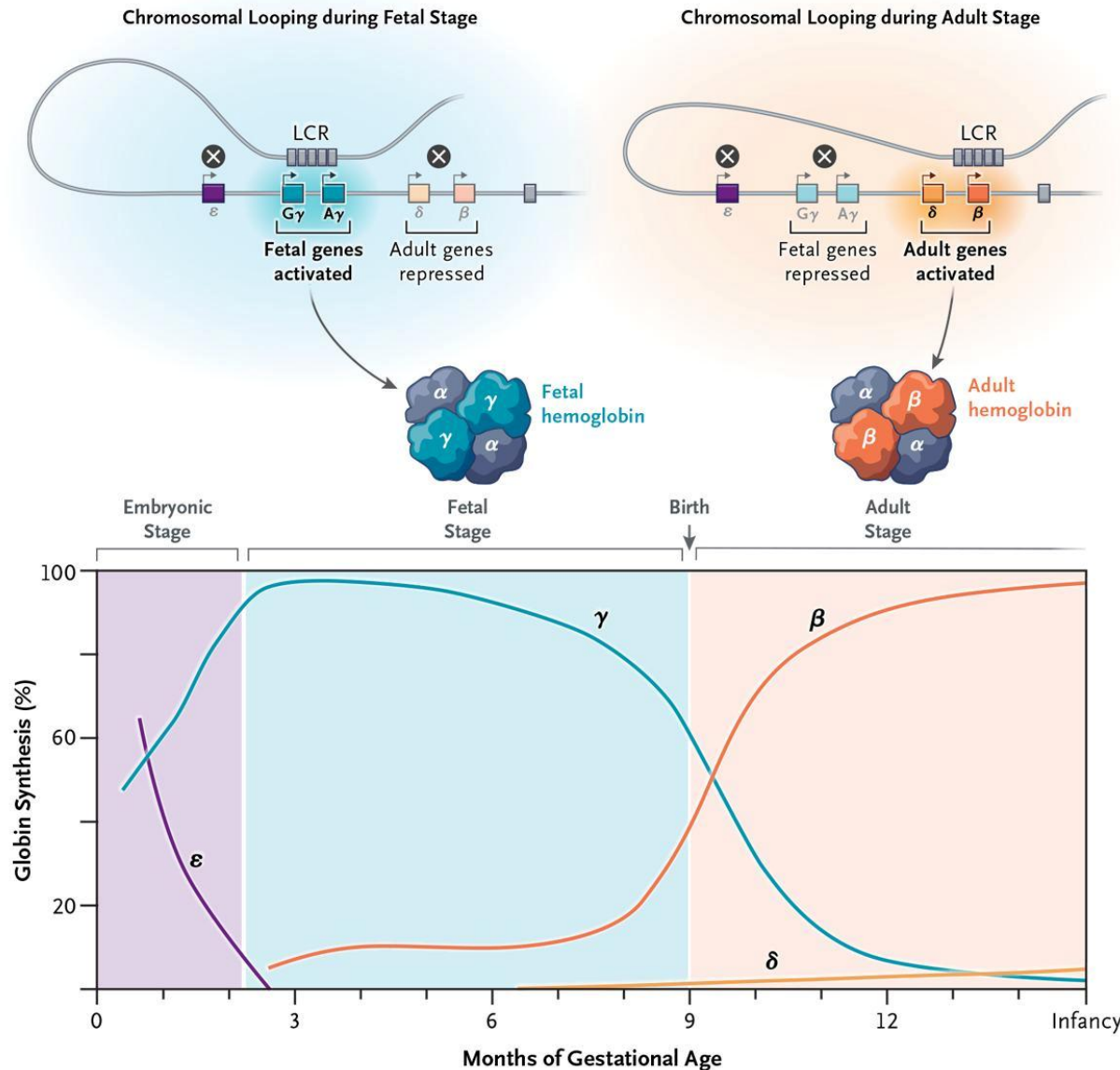


The Fetal-to-Adult Hemoglobin Switch — Mechanism and Therapy



Locus Control Region (LCR)

The Fetal-to-Adult Hemoglobin Switch — Mechanism and Therapy

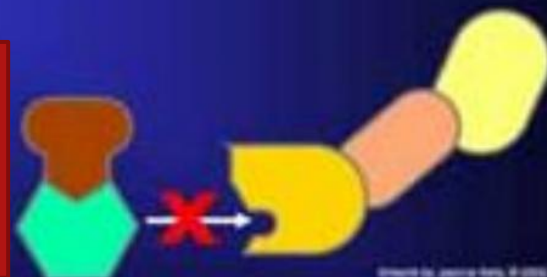


Locus Control Region (LCR)



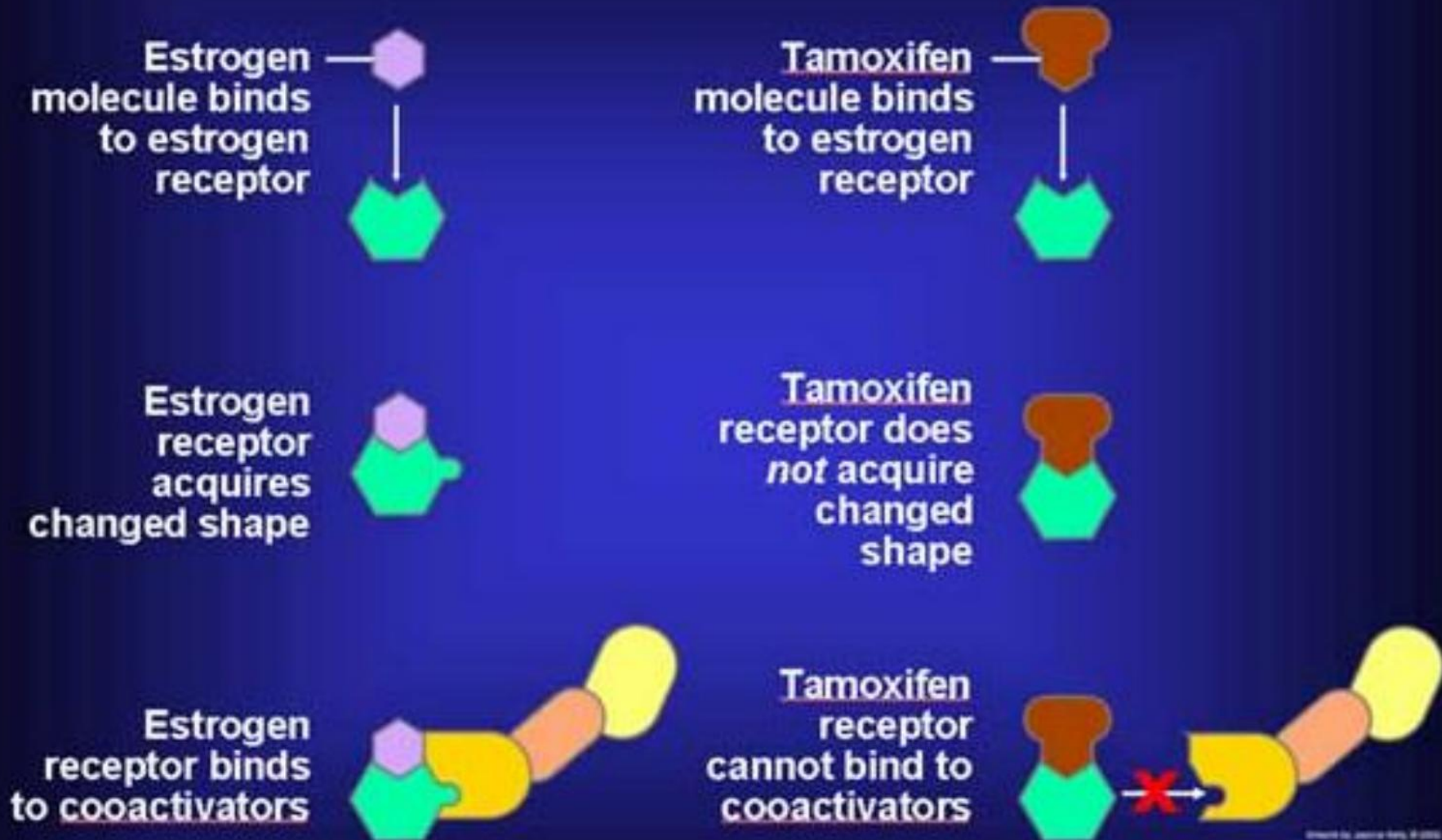
a)

c)



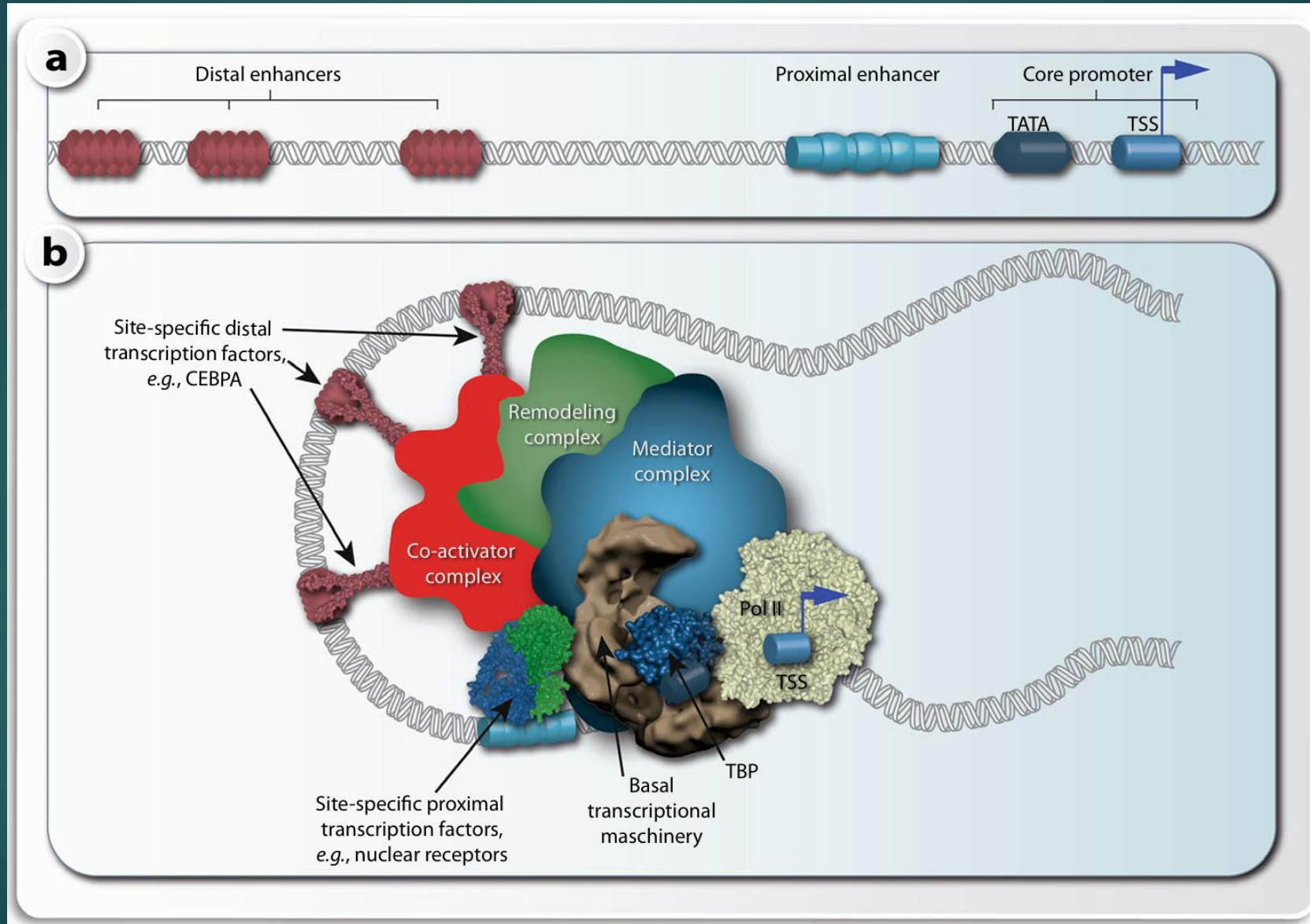
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Tamoxifen and Cancer

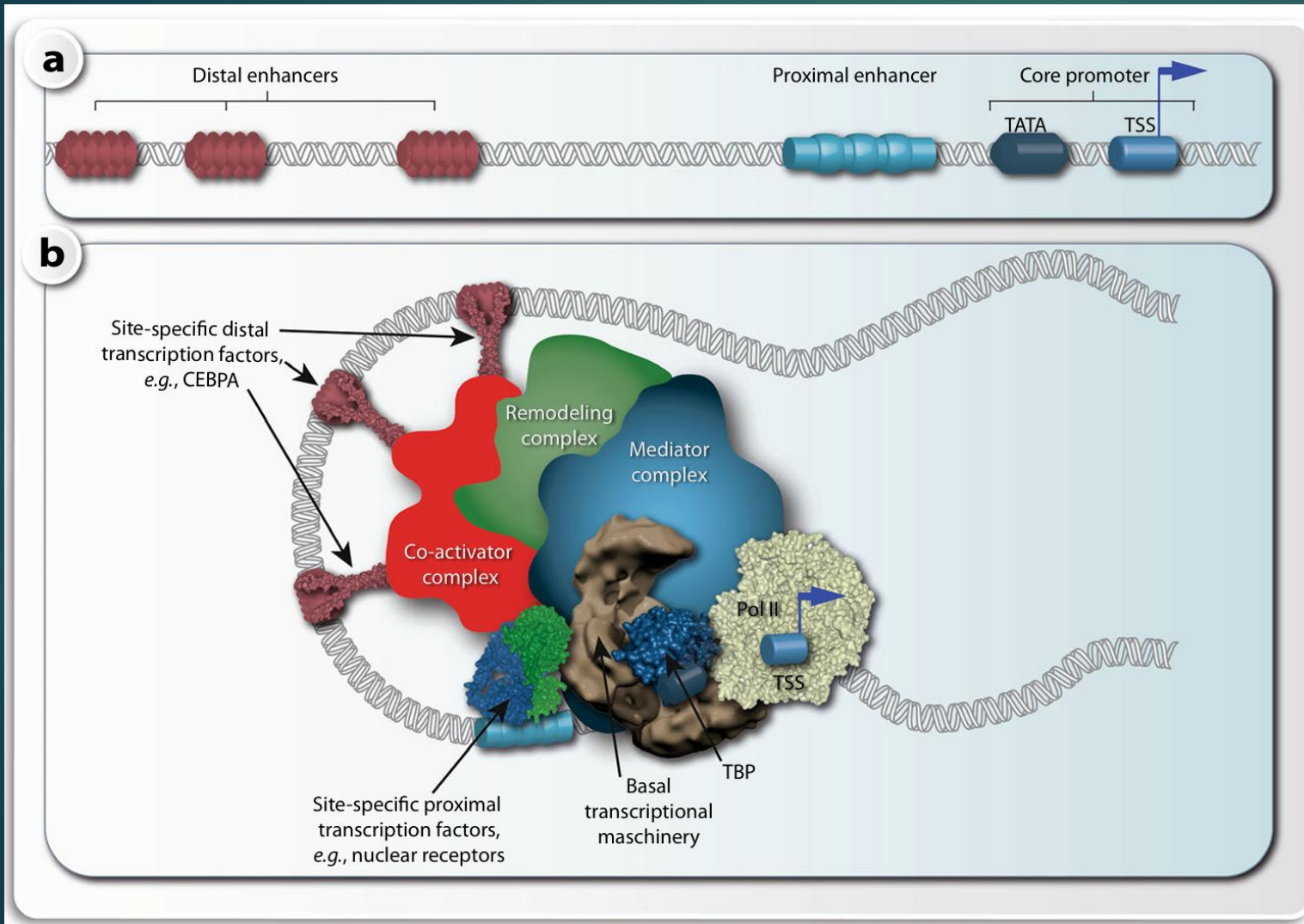


Interpretation

Components of transcriptional regulation



Components of transcriptional regulation

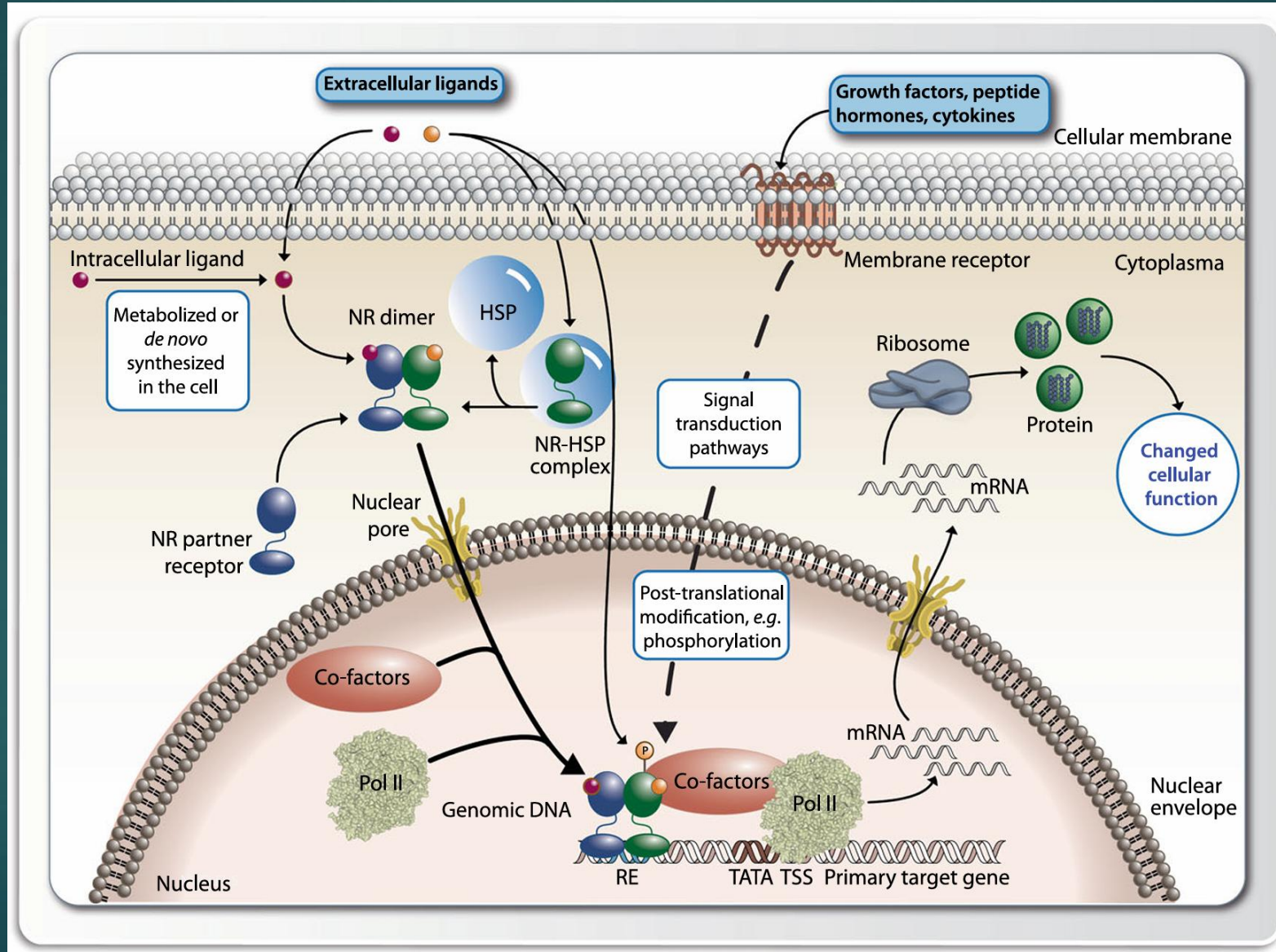


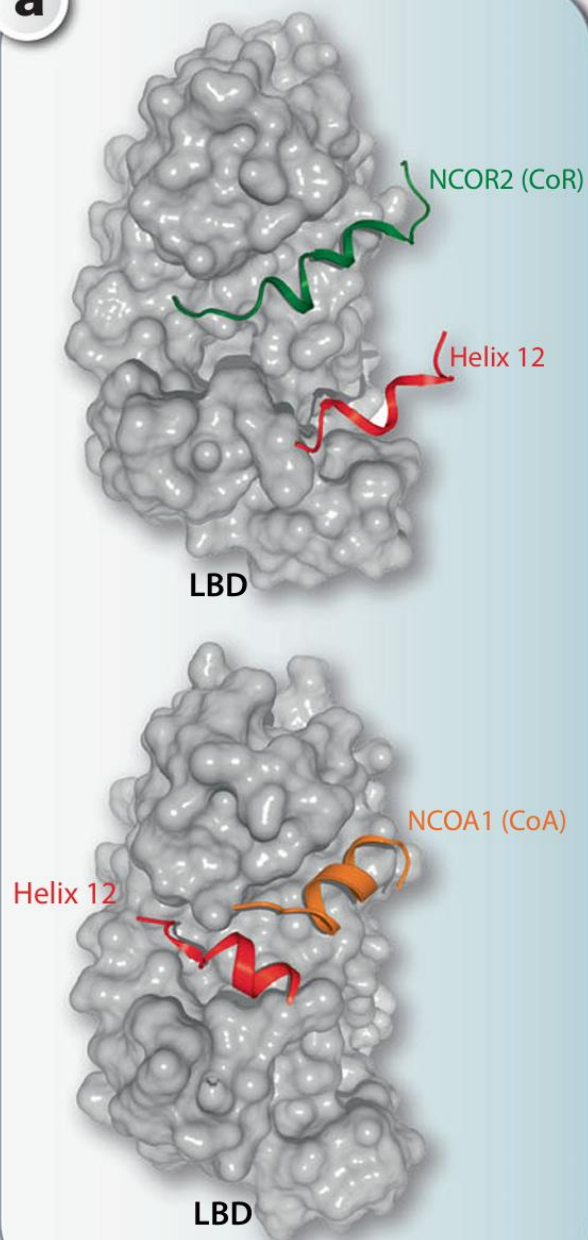
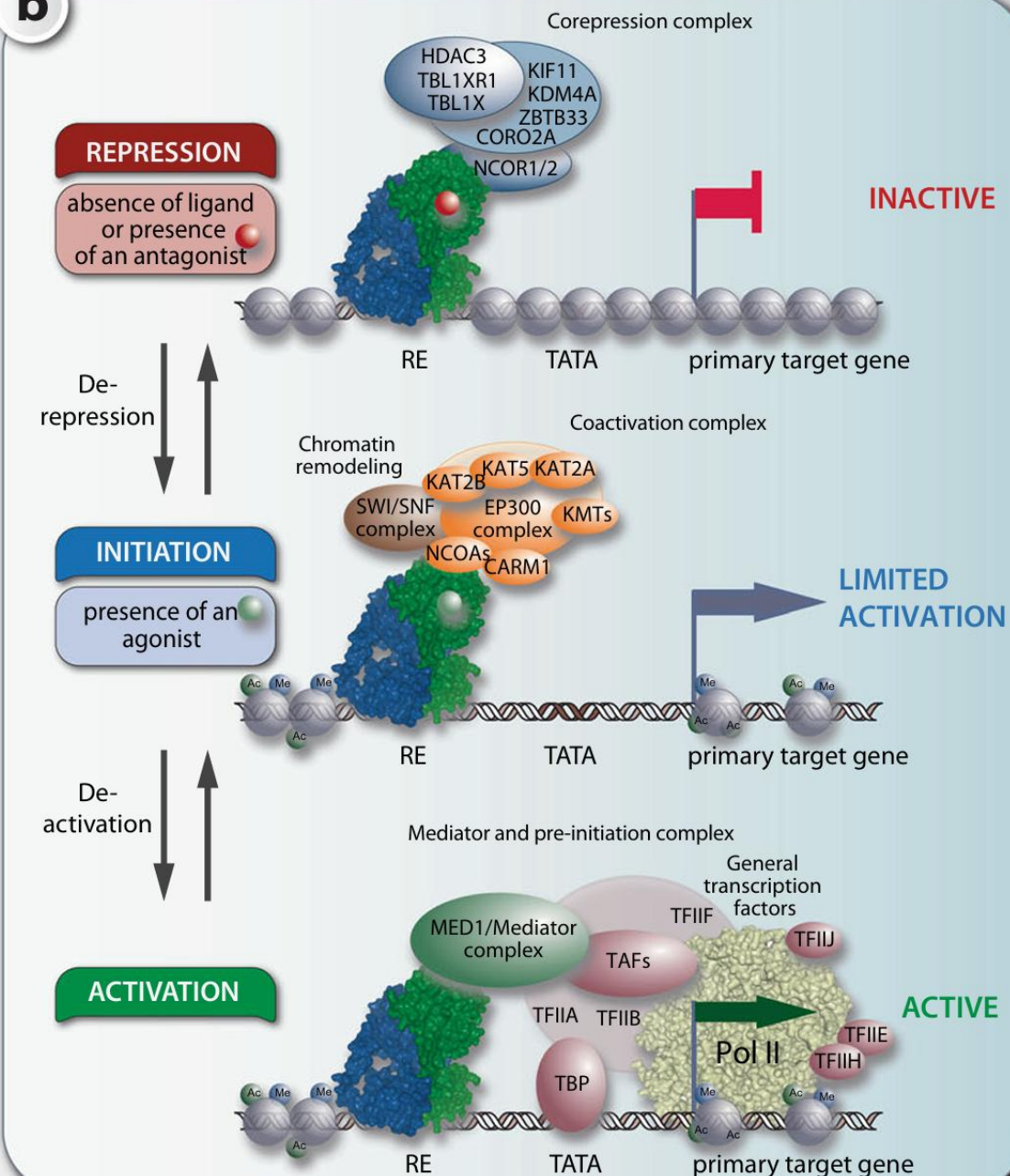
Basal is Insufficient: As noted, the GTFs + RNAPII alone are inefficient *in vivo* due to chromatin and the need for precise signals.

The Bridge: The Mediator Complex

Significance: Mediator integrates signals from multiple regulatory proteins and transmits them to the basal machinery to stabilize the PIC and greatly stimulate transcription rates.

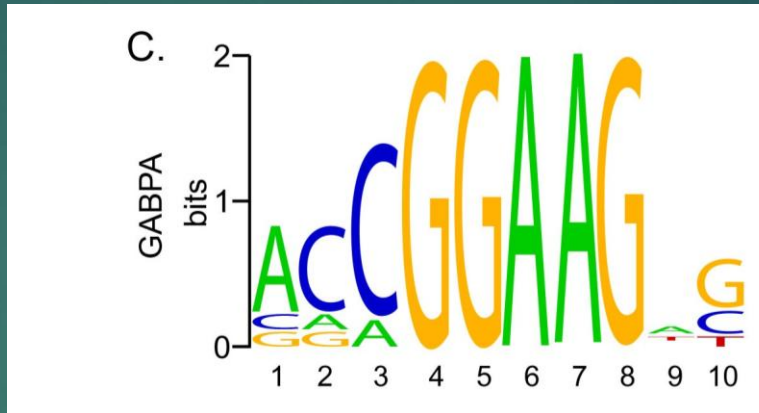
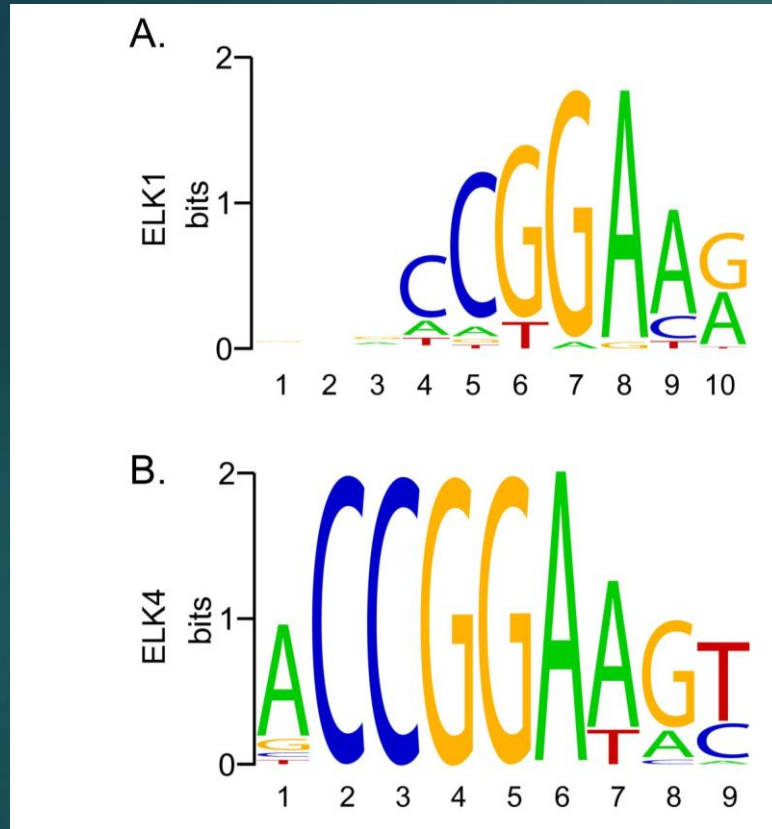
Principles of nuclear receptor signaling



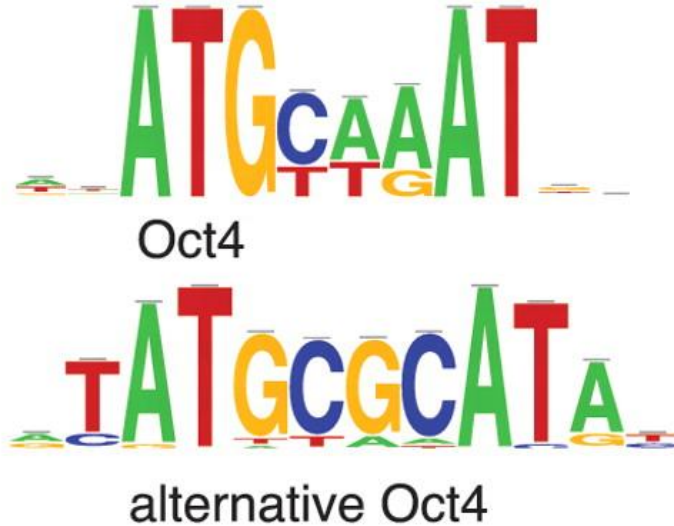
a**b**

Interaction of nuclear receptors with cofactors. A solvent excluded surface (Connolly surface) representation of a nuclear receptor LBD in the absence (top) and presence (bottom) of a ligand

Transcription Factor Binding Sites Motifs

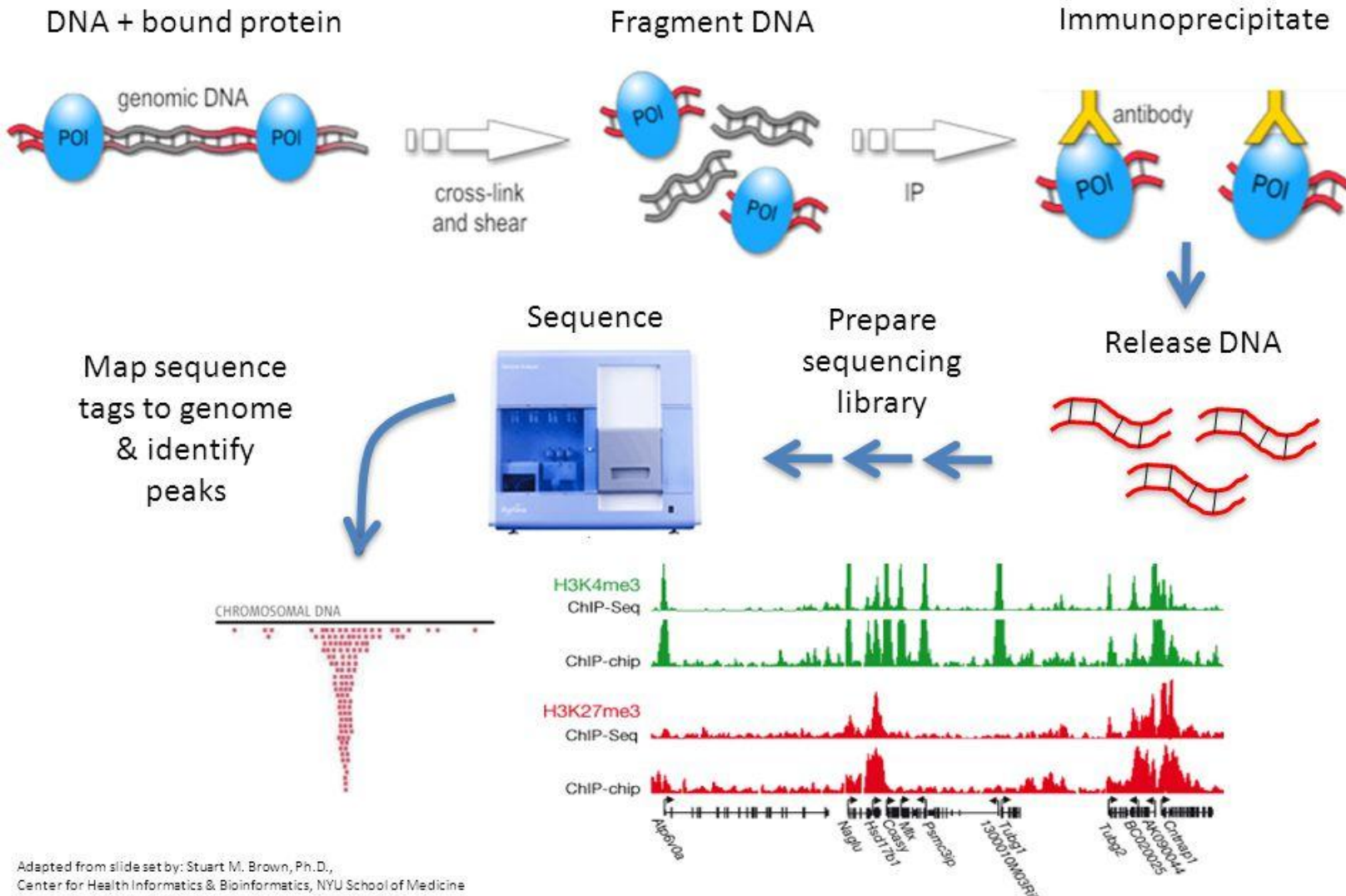


Sequence logos of ETS transcription factor binding sites. Sequence logos of consensus DNA binding sites for the three ETS transcription factors enriched in genes regulated by HNF4a. Y axis indicates amount of information at each position in the motif. These logos were generated from information obtained from the JASPAR core database [42]. (A), ELK1 binding motif; (B), ELK4 binding motif; and (C), GABPA binding motif.



In the Sox2 and Oct4 peak sets, peak-motifs found not only the composite 'SOCT' motif bound by the Sox2/Oct4 complex (reported by Chen and co-workers), but also the distinct motifs recognized by Sox2 (CTCTTTGTT) and Oct4 (ATGyAAAt), respectively (Figure 4 , top). Interestingly, in the Oct4 data set, unknown motifs were returned with a high significance, (i.e. motifs with no significant similarity with the consensus encompassed by the common databases). Such motifs may reveal alternative consensus, as in the case of the motif crTATGCGCATAyG, which actually corresponds to an alternative Oct4 motif, also detected in other recent studies

ChIP-seq overview



Adapted from slide set by: Stuart M. Brown, Ph.D.,
Center for Health Informatics & Bioinformatics, NYU School of Medicine

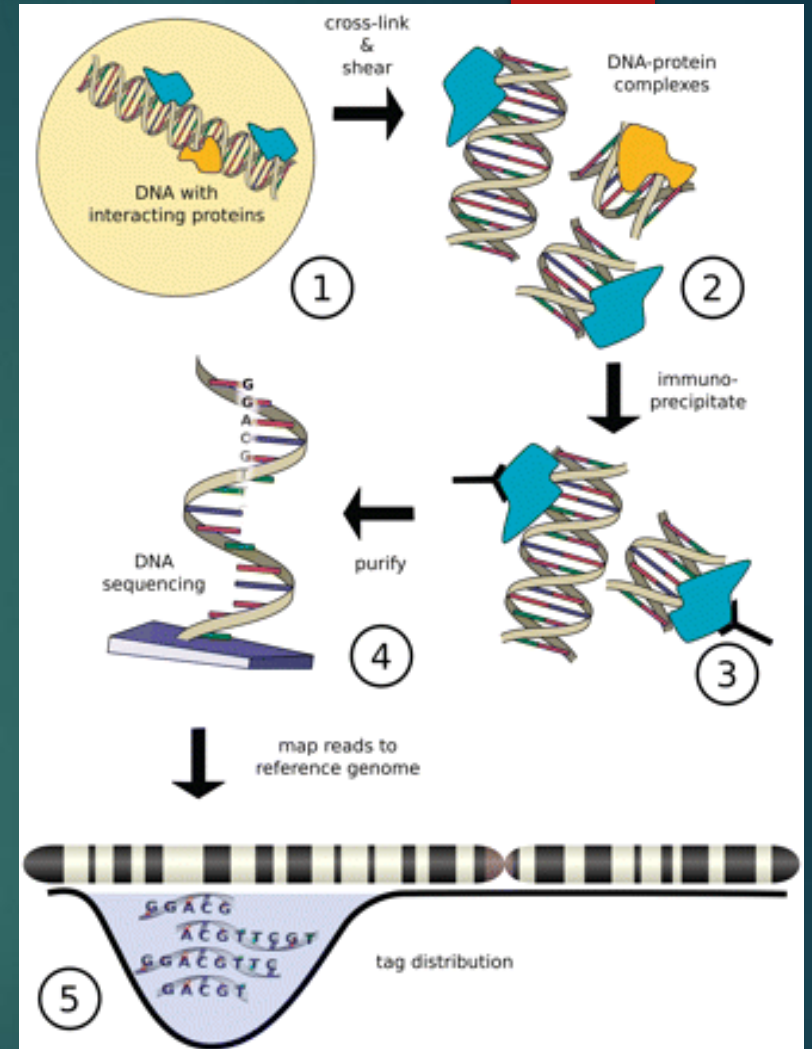
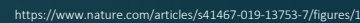
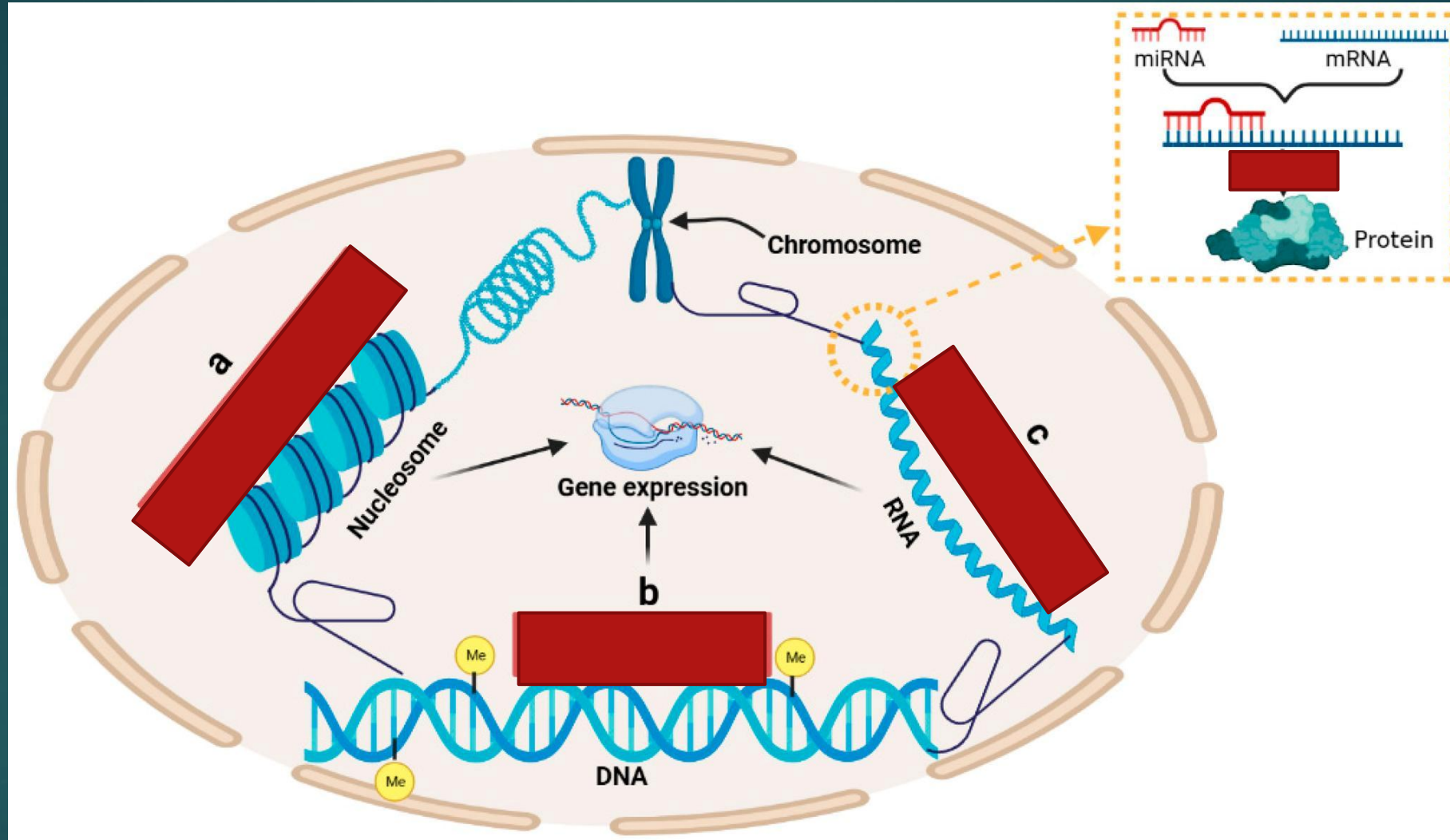


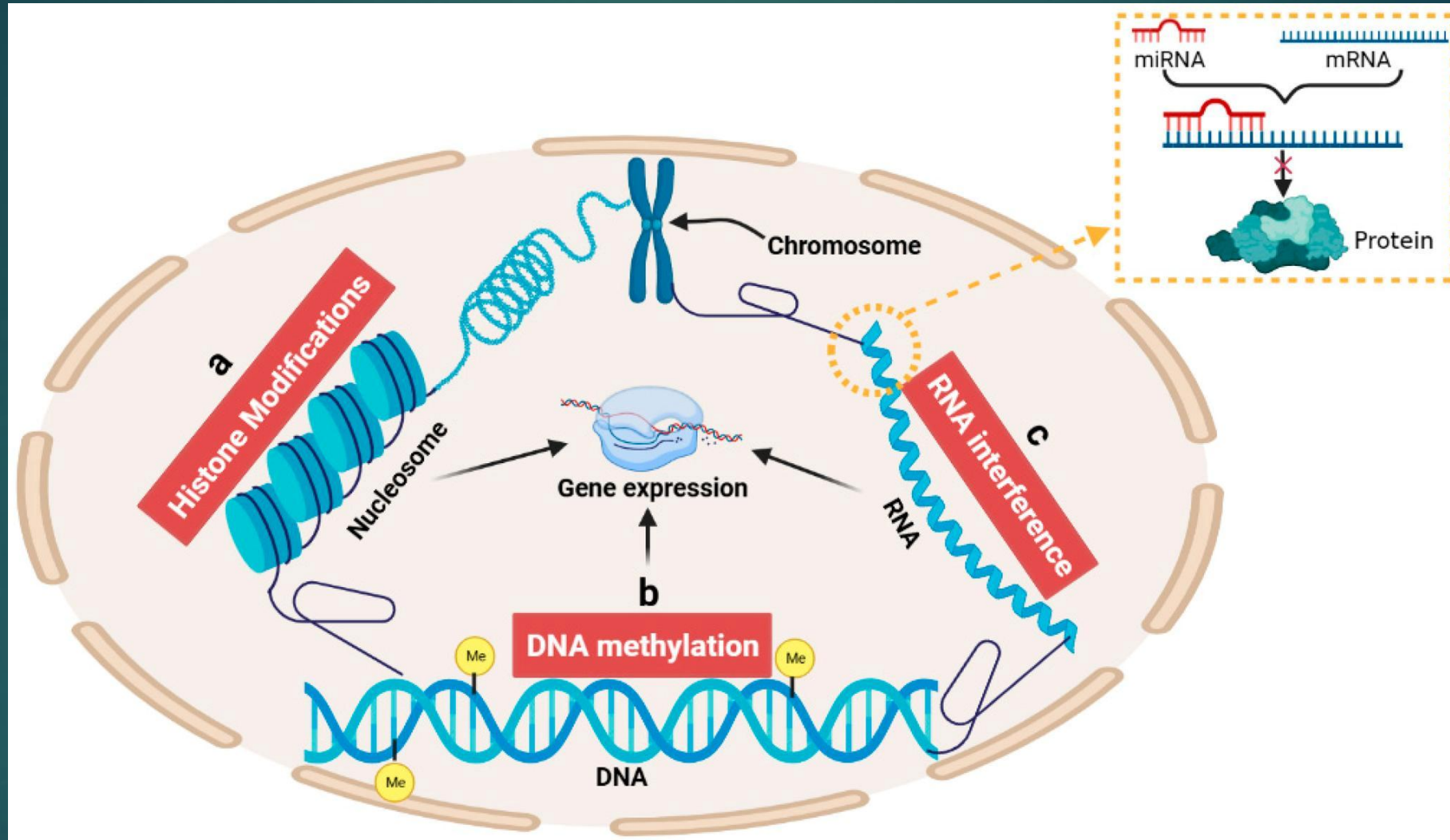
Figure from Szalkowski, A.M, and Schmid, C.D.(2010). *Rapid innovation in ChIP-seq peak-calling algorithms is outdistancing benchmarking efforts*. Briefings in Bioinformatics.



Epigenetic Mechanisms in Gene Regulation



Epigenetic Mechanisms in Gene Regulation





Thank you for your time!

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